

Loan Default Prediction

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1. Business context:

Lending Club is a peer-to-peer lending platform that provides loans to individual borrowers.

One of the main business risks is loan default, where borrowers fail to repay their loans.

High default rates lead to financial losses and poor risk pricing.

Therefore, predicting loan default at the time of loan origination is critical for improving credit decisions and reducing losses.

2. Stakeholders

The primary users of this predictive model are:

- Credit risk analysts who assess borrower risk levels
- Loan approval teams responsible for accepting or rejecting applications
- Risk management departments that monitor portfolio performance

These stakeholders rely on the model's predictions to make informed lending decisions and reduce exposure to high-risk borrowers.

3. Business Decision

The predictive model supports key business decisions, including:

- Approving or rejecting loan applications based on risk level
- Adjusting interest rates to reflect borrower risk
- Flagging high-risk loans for additional manual review

By integrating these predictions into the lending process, the company can reduce default rates

while maintaining profitable lending operations.

4. Success Metrics

Loan default prediction is an imbalanced classification problem, where default cases represent a minority of the dataset.

Therefore, accuracy alone is not sufficient to evaluate model performance.

The main evaluation metrics used are:

- ROC-AUC to measure the model's ability to distinguish between defaulters and non-defaulters
- Precision-Recall AUC to assess performance on the minority default class
- Recall for default cases to minimize missed high-risk borrowers

These metrics provide a more reliable assessment of model effectiveness in real-world risk scenarios.

5. Constraints and Risks

Several challenges were identified during the project:

- The target variable is highly imbalanced, with default cases representing a small proportion of the dataset
- The dataset contains missing values and noisy real-world financial data
- Some features contain post-loan information, which can cause data leakage if not removed

6. Project Objective

The objectives of this project are to:

- Build a clean, analysis-ready and leakage-free dataset
- Perform thorough data quality assessment and exploratory analysis
- Train and compare multiple machine learning models
- Identify the key drivers of loan default

7. Model Evaluation & Explainability

Model Performance:

The final models were evaluated using ROC-AUC and Precision-Recall AUC metrics.

After removing data leakage features and applying proper preprocessing, the logistic regression model achieved stable and realistic performance, with strong discrimination between default and non-default loans.

Model Explainability

Feature importance analysis using logistic regression coefficients was conducted to understand key drivers of loan default.

8. Modeling Choices, Trade-offs

Data Leakage Handling:

Several post-loan features related to payments, recoveries, and hardship events were identified and removed to prevent unrealistic model performance.

Although removing these features reduced accuracy, it resulted in a reliable and deployable model.

Missing Value Strategy:

Different imputation techniques were tested, including median imputation and regression-based imputation.

Median imputation was selected due to similar or better performance with lower complexity and higher stability.

Model Selection Trade-offs:

Logistic Regression was chosen over Random Forest because:

- Slightly better performance
- Higher interpretability
- Lower risk of overfitting